

Enhanced digestibility and fermentability of sugarcane bagasse in biofuel production by surfactant-assisted dilute acid pretreatment

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ABSTRACT

In this study, various combined pretreatment strategies using liquid hot water (LHW), 60 % ethanol, 1% H₂SO₄ and 0.5 % NaOH together with different surfactants (PEG 6000, Triton X-100 and Tween 80) were evaluated to improve glucose production from sugarcane bagasse (SCB) during enzymatic hydrolysis. After pretreatment with PEG 6000-assisted 1% H₂SO₄, the highest glucose yield of 75.04 % was achieved. Pretreatments of SCB with various combinations of H₂SO₄ (1–3%, w/v) and PEG 6000 (0–1.5 %, w/v) were further investigated for enhancing enzymatic hydrolysis and ethanol production. When 1.5 % PEG 6000 was introduced during pretreatment with 2 % H₂SO₄, the highest glucose yield of 86.14 % and ethanol concentration of 20.17 g/L were liberated, which were much higher than those obtained from pretreatment with 2% H₂SO₄ alone. This increased digestibility and fermentability can be attributed to decreases in crystalline sizes and PEG 6000 adherence to SCB surfaces, which was confirmed by cellulase adsorption isotherms, FT-IR, XRD, and SEM analysis. In addition, prehydrolysate and the stillage remaining after ethanol production were sequentially digested to seek potential biomethane production and resulted in final methane yields of 264.05 mL and 387.77 mL per g of volatile solids, respectively. The feasibility of the process provides an environmentally friendly technology for an integrated biorefinery.

1. Introduction

Due to dwindling reserves of fossil fuels worldwide, lignocellulosic biofuels, which are renewable and available resources, are desirable alternatives for easing energy limitations. Sugarcane bagasse is one of the most abundant lignocellulosic raw materials in southern China, and its abundant carbohydrate content allows conversion into renewable products (Zhang et al., 2019a). However, recalcitrant structures containing cellulose, hemicellulose and lignin strongly impede efficient conversion (Bansal et al., 2016; Khazraie et al., 2017). Hence, pretreatment was proposed to deconstruct the complex matrix structure, increase enzyme accessibility and liberate more fermentable sugars (Vats and Negi, 2013). Recently, various pretreatment technologies (such as LHW, steam explosion, dilute acid/alkali, organosolvents and ionic liquid pretreatment) have been widely investigated (Bhatia et al., 2019; Meneses et al., 2020; Zhang et al., 2019b). However, some lignin reprecipitate on the lignocellulosic surface after those pretreatments, especially those involving dilute acid, dilute alkali and organosolvents; these precipitates have a high affinity for cellulolytic enzymes and

decrease the efficiency of hydrolysis and fermentation (Pandey and Negi, 2015). To avoid this phenomenon, a combined surfactant pretreatment has been proposed because surfactants decrease liquid surface tension and prevent redeposition of hydrophobic compounds on biomass, which would otherwise result from the hydrophilic and hydrophobic properties of the surfaces (Qing et al., 2010).

Surfactants have been shown to adsorb into lignin and prevent unproductive adsorption of enzymes (Camila and Johanna, 2020). Several reports of surfactant-assisted pretreatment have indicated the effectiveness of this approach in enzymatic saccharification and ethanol fermentation. According to the literature, the improved enzymatic conversion after surfactant addition arises from increased accessible surface area of cellulose (Nasirpour and Mousavi, 2018), removal or modification of the lignin structure (Wang et al., 2020a), and stabilization of enzyme activity, especially at high temperatures (Eckard et al., 2013). Qi et al. suggested that surfactant has little influence on chemical composition but modifies the surface of lignin to improve enzymatic hydrolysis (Qi et al., 2010a). Sindhu et al. found that Triton X-100-assisted acid pretreatment of sugarcane tops effectively removed

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Table 1Chemical composition of raw material and PEG 6000 assisted H₂SO₄ pretreated SCB with different concentration.

Sample	Concentration (%)		Solid recovery (%)	Contents (%)					
	H ₂ SO ₄ (w/v)	PEG 6000 (w/v)		Glucan	Recovery	Xylan	Removal	AIL	Removal
	Raw material		–	44.21		23.88		21.24	
1		0	73.13	57.64	95.34	7.96	75.62	28.49	1.91
2	1 %	0.5 %	73.17	57.79	95.65	8.21	74.84	28.73	1.03
3		1.0 %	73.21	57.75	95.63	8.38	74.31	28.41	2.07
4		1.5 %	73.30	57.56	95.43	8.13	75.04	28.32	2.27
5		0	69.12	60.09	93.95	6.99	79.77	29.36	4.46
6	2 %	0.5 %	69.36	60.01	94.15	6.81	80.22	29.65	3.18
7		1.0 %	69.13	60.36	94.38	6.61	80.86	29.96	2.49
8		1.5 %	69.46	60.15	94.50	6.77	80.31	29.56	3.33
9		0	67.20	61.05	92.80	5.41	84.78	30.31	4.10
10	3 %	0.5 %	67.27	61.23	93.17	5.25	85.21	30.65	2.93
11		1.0 %	67.3	61.09	92.99	5.38	84.84	30.41	3.65
12		1.5 %	67.35	61.01	92.94	5.33	84.96	30.23	4.14

lignin and yielded high proportions of reducing sugars (Sindhu et al., 2012). Similarly, PEG-assisted ionic liquid pretreatment of biomass improved bioethanol yield by decreasing the content of hemicellulose and lignin (Goshadrou and Lefsrud, 2017; Nasirpour and Mousavi, 2018). Furthermore, Pandey & Negi reported that surfactant-assisted acid pretreatment was more economical than surfactant-assisted base pretreatment for ethanol production (Pandey and Negi, 2015). However, there are few reports of comparisons among surfactants-assisted LHW/ethanol/acid/alkali pretreatments.

Simultaneous saccharification and fermentation (SSF) generated ethanol by minimizing the negative effects of fermentable sugars on cellulase (Fan et al., 2018). Prior to ethanol generation, carbohydrate polymers (cellulose and hemicellulose) were hydrolyzed into monomers (hexoses and pentoses). However, common industrial yeast strains are unable to efficiently metabolize pentoses, which remain as biomass waste in the stillage or hydrolysate. To improve the efficiencies of such processes, conversion of residual sugar by anaerobic digestion was proposed as an environmentally friendly and feasible strategy (Divya et al., 2015; Fuess and Garcia, 2014; Kalyani et al., 2017). Recently, there have been several studies on biogas production from prehydrolysates or fermentation stillages (Liu et al., 2015; Patel, 2017; Wang et al., 2018a). A high methane production of 318.2 mL/g COD from hydrolysates of mild acid pretreated rice straw was reported by (Wang et al., 2018a). Katsimpouras et al. evaluated methane production from fermentation residues after ethanol production from steam-exploded corn stover with 0.2 % H₂SO₄, and the residues exhibited a final methane yield of 120 mL/g VS (Katsimpouras et al., 2017). However, most studies have been conducted for separate and individual experimental stages, and there have been few investigations of combined digestion.

In this study, surfactants (PEG 6000, Triton X-100 and Tween 80)-assisted LHW, organosolve (ethanol), 1% H₂SO₄ and 0.5 % NaOH pretreatments of SCB were explored by assessing glucose yields from enzymatic hydrolysis. Then, the influence of PEG 6000 and H₂SO₄ loadings on SCB pretreatment was determined from structural composition changes and hydrolysis efficiency. The relevant features of raw and pretreated SCB were characterized with cellulase adsorption isotherms, Fourier transform infrared (FTIR), X-ray diffraction (XRD) and scanning electron microscopy (SEM). Furthermore, SSF treatments of substrates exhibiting high recoveries of total sugar (glucose and xylose) were used to produce bioethanol. Specifically, stillage and prehydrolysate (mainly xylose) were subsequently subjected to biomethane generation to seek a cleaner production method to reduce environmental pollution and improve the yield of utilized biomass.

2. Materials and methods

2.1. Materials

Chipped SCB was collected from Shaoguan, China. The dried material was preliminarily treated by milling and screening (<1 mm) and then stored at room temperature until use. The chemical composition of the SCB was determined according to a previous method (Sluiter et al., 2008), and the results are described in Table 1. The raw SCB contained 44.21 % glucan, 23.88 % xylan and 21.24 % acid insoluble lignin (AIL).

Three types of non-ionic surfactants: PEG 6000, Triton X-100 and Tween 80 were all purchased from Aladdin Industrial Corporation.

Cellulase (Cellic CTec2) used in the enzymatic hydrolysis and SSF was measured to have a filter paper activity of 164 FPU/mL (Adney and Baker, 1996).

The *Saccharomyces cerevisiae* as a commercial yeast was obtained from Angel Yeast Co., Ltd. (Yichang, China).

2.2. Pretreatment method

It has been reported that different pretreatment conditions can damage the structure of lignocellulose to varying degrees, especially thermochemical pretreatments (Wang et al., 2020b). Typical thermochemical pretreatment methods, including those using LHW, dilute acid, dilute acid and organosolve pretreatment have been improved (Meng et al., 2015). During the selection of different types of surfactant (including nonionic, anionic, cationic and biosurfactants), nonionic surfactants were favored for drastically enhancing the enzymatic conversion of lignocellulosic biomass into fermentable sugars (Kristensen et al., 2007). Therefore, to evaluate the influence of surfactants on pretreatment and subsequent enzymatic hydrolysis, four representative pretreatment strategies using LHW/60 % ethanol/dilute acid/dilute alkali with or without surfactant were conducted in a steam sterilizer at 121 °C for 60 min (Zhang et al., 2018). All pretreatments were performed in a 100 mL pressure bottle (Synthware, Beijing, China) with a solid-to-liquid ratio of 1:10. After pretreatment, the solids were filtered and washed using deionized water until a neutral pH was reached. Liquid fractions were stored in a 4 °C refrigerator for further sugar analysis (Sluiter et al., 2008). Pretreated SCB was stored in a refrigerator for subsequent enzymatic hydrolysis and fermentation.

2.3. Enzymatic hydrolysis of SCB

Pretreated SCB was enzymatically hydrolyzed with a cellulase dose of 20 FPU/g dry substrate. Enzymatic digestion experiments were carried out in pH 4.8 acetate buffer with a substrate concentration of 2% (w/v) at 50 °C, and solutions were stirred at 150 rpm for 72 h. During enzymatic hydrolysis, 1 mL of supernatant was withdrawn at 6, 12, 24,

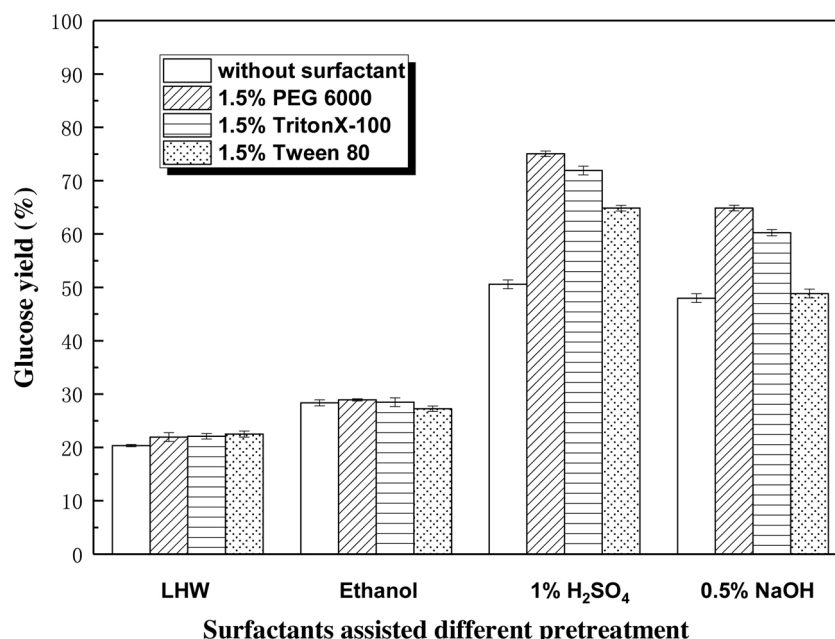


Fig. 1. Glucose yield of different surfactants assisted LHW/ethanol/1% H₂SO₄/0.5 % NaOH pretreatment under enzymatic hydrolysis time of 72 h.

48, and 72 h for analysis of xylose and glucose by HPLC.

The sugar contents in the structural composition, prehydrolysate and enzymatic hydrolysate were detected by HPLC (Shimadzu, Japan) with a refractive index (RI) detector. Separation was conducted on a SUGAR KS-801 cation-exchange column at 60 °C with a mobile phase of ultra-pure water and a flow rate of 0.4 mL/min.

Inhibitors (acetic acid, HMF and furfural) were quantified by HPLC with prominence UV/VIS detector (SPD) and a SUPELCOGEL C-610H column at 35 °C with 0.5 mL/min of 0.1 % H₃PO₄.

2.4. Fermentation

The SSF of pretreated SCB was conducted in a 250 mL flask at a pH of 4.5. Before SSF, yeast was preliminarily activated using the method described in a previous report (Fan et al., 2018). Then, 10 g of dry solid substrate, 5 mL of activated yeast (3.3 g of cell concentrate), enzyme solution (15 FPU/g pretreated SCB) and a certain volume of sterile distilled water were mixed together to achieve a final liquor volume of 100 mL and incubated at 34 °C with stirring at 150 rpm for 72 h. All experimental procedures were run under sterile conditions. Supernatant samples were collected at 4, 8, 12, 24, 48, and 72 h and studied by HPLC. HPLC was performed at 50 °C to quantify ethanol concentrations on a SUGAR SH1011 cation-exchange column coupled to a refractive index detector (RID). A mobile phase comprising 0.05 M H₂SO₄ was used with a flow rate of 1.0 mL/min.

2.5. Biomethane production

The prehydrolysates obtained from pretreatment and the remaining residues after fermentation were followingly conducted to BMP experiments. A batch assay was carried out in sealed 500 mL serum bottles at 37 °C under a 1:2 (substrates: inoculum) based on VS mixture ratio until no gas determined. The inoculum was obtained from the Datansha Sewage treatment plant in Guangzhou, China and cultivated in a mesophilic anaerobic fermentation tank in our laboratory over a long period. The characterization of substrates used in the BMP experiments were described in Table 3. Specific experimental procedures were described in our previous literature (Fan et al., 2019).

2.6. Analytical methods

The yield of glucose and ethanol obtained from enzymatic digestibility and SSF were calculated using Eqs. (1) and (2), respectively:

$$\text{Glucose yield (\%)} = \frac{\text{Glucose production in enzymatic hydrolysis}}{\text{Glucan amount in raw material} \times 1.11} \times 100\% \quad (1)$$

$$\text{Ethanol yield (\%)} = \frac{\text{Ethanol produced (g)}}{\text{cellulose content (g) in pretreated SCB} \times 0.568} \times 100\% \quad (2)$$

2.7. Characterization analysis of the samples

The crystallinity index (CrI) and crystallites size (D) of samples were analyzed with X-ray diffraction using a Bruker D8-ADVANCE (Karlsruhe, Germany), and this radiation was Cu K α ($\lambda = 1.54 \text{ \AA}$) in grade ranges between 5 ° to 60 °. Their CrI and cellulose crystallites size (D) were calculated according to the Segal method (Segal et al., 1959) and Scherrer equation (Kumar et al., 2009), respectively.

The morphology of raw and pretreated SCB were observed by scanning electron microscopy (EVO18, ZEISS, Germany).

Fourier transform infrared spectroscopy were detected using a PerkinElmer spectrometer. 1.5 mg of dried sample was embedded in 100 mg of spectroscopic grade KBr pellets. The spectra were performed in an average of 25 scans and a resolution of 4 cm⁻¹, in the range of 4,000–400 cm⁻¹.

2.8. Cellulase adsorption isotherm

The cellulase adsorption isotherm for the different samples was measured by incubating substrates (2%, w/v) with a range of cellulase concentration (0.1–2 mg/mL) in 50 mM acetate buffer (pH 4.8), following the method of (Sheng et al., 2017).

2.9. Statistical analysis

Significance about the results of the chemical composition and enzymatic hydrolysis was analyzed and set when p value <0.05.

Table 2

Sugar and inhibitor in the hydrolysate during various surfactant-assisted dilute acid pretreated SCB.

Concentration (%)		Sugar analysis (g/100 raw materials)				Inhibitor (g/L)		
H ₂ SO ₄ (w/v)	PEG 6000 (w/v)	Total glucose	Glucose Mono.	Total xylose	Xylose Mono.	Acetic acid	Furfural	HMF
1%	0	1.95	1.18	18.61	17.4	3.57	0.11	0.02
	0.5 %	1.94	1.20	18.87	16.91	3.64	0.11	0.02
	1.0 %	2.00	1.21	19.16	16.89	3.64	0.11	0.02
	1.5 %	1.92	1.14	18.57	16.42	3.67	0.11	0.02
2%	0	2.21	1.91	20.31	19.49	4.04	0.31	0.03
	0.5 %	2.23	1.93	20.79	19.09	3.91	0.33	0.03
	1.0 %	2.23	1.92	20.61	18.83	3.86	0.35	0.03
	1.5 %	2.20	1.87	20.88	18.35	3.80	0.33	0.03
3%	0	2.45	2.23	20.21	19.49	4.10	0.59	0.04
	0.5 %	2.47	2.23	20.78	19.09	4.01	0.63	0.04
	1.0 %	2.43	2.21	20.87	19.45	4.07	0.59	0.04
	1.5 %	2.53	2.32	21.13	19.82	3.97	0.60	0.04

3. Results and discussion

3.1. Comparison of the effectiveness of surfactants-assisted LHW/acid/alkali/ethanol pretreatments

It has been reported that surfactants-assisted pretreatment effectively enhanced the saccharification of lignocellulose (Chang et al., 2016; Qing et al., 2010). In this section, three typical surfactants involving PEG 6000, Triton X-100 and Tween 80 with identical 1.5 % (w/v) doses were screened to investigate their impacts on LHW (a), 60 % ethanol (b), 1% H₂SO₄ (c) and 0.5 % NaOH (d) pretreatments, and the results are presented in Fig. 1. After pretreatments with LHW or ethanol (Zhang et al., 2018), glucose yields without surfactant were 20.35 % and 28.36 %, respectively. There was no significant difference in glucose yield with surfactant addition. Although more than 95 % solids were recovered with or without surfactant addition, the relatively low glucose yield indicated that the intact structure of the SCB was only slightly disrupted under mild reaction conditions and that the addition of surfactants did little to improve the glucose yield (Wei et al., 2017). However, for the 1% H₂SO₄ and 0.5 % NaOH pretreatments, an obvious increase in glucose yield was observed with surfactant addition. Among the surfactants, PEG 6000 was found to be more effective in liberating glucose, and glucose yields reached 75.04 % and 64.88 %, corresponding to increase of 48.35 % and 35.21 %, after acid and alkali pretreatments, respectively. Although additions of Triton X-100 and Tween 80 also enhanced glucose yield, their glucose yields and increases were much lower than that those seen with PEG 6000. PEG 6000-assisted 1% H₂SO₄ pretreatment presented the finest performance in improving glucose yield; hence, PEG 6000 was used with H₂SO₄ pretreatment to facilitate enzymatic hydrolysis of SCB.

3.2. Effect of PEG 6000-assisted H₂SO₄ pretreatment on SCB chemical composition

PEG 6000 (0–1.5 %, w/v)-assisted H₂SO₄ (1–3%, w/v) pretreatments were carried out to investigate their effects on enzymatic hydrolysis. Total solids recoveries and SCB compositional changes seen with various pretreatments are summarized in Table 1. Native SCB contained 44.21 % cellulose, 23.88 % hemicellulose and 21.24 % acid-insoluble lignin (AIL). As shown in Table 1, when the H₂SO₄ concentration was held constant, changes in total gravimetric recovery and proportions of macro components (cellulose, hemicellulose, and lignin) did not vary with increasing concentrations of PEG 6000. In contrast to the effect of PEG 6000, the impact of H₂SO₄ concentration on the chemical composition of pretreated solids was more remarkable. As the acid concentration was increased from 1% to 3% in the absence of PEG 6000, the solid loss gradually increased from 26.87 % to 32.80 %. This was attributed to larger degrees of hemicellulose solubilization occurring

Table 3

Characterization of substrates used in the BMP experiments after 9 days.

parameter	Hydrolysate	SCB residues	Inoculum
pH	0.71	4.50	8.1–8.5
TS (%)	3.61 %	19.11 %	2.33 %
VS (%)	3.73 %	18.38 %	1.65 %
COD (mg/L)	5700	N.D	680 g/L
Feed ratio	1:2	1:2	–
(substrate: inoculum)			
Biomethane production (ml/g vs)	378.65	264.05	–

N.D. non determined.

with higher H₂SO₄ concentrations. The increased loss of xylan (from 77.51%–85.95 %) further suggested removal of a large proportion of hemicellulose and depolymerization upon addition of acid alone.

Furthermore, although there was 44.21 % of glucan in the raw material, the glucan content in pretreated SCB ranged from 57.56%–61.23%. The increase in glucan content was consistent with the reduction in xylan content seen with pretreated SCB. AIL contents increased from 28.32 % to 30.85 %, suggesting that approximately 1.59 %–4.14 % of the lignin in the feedstock was removed. Although some studies have reported that surfactants during pretreatment increased the solubility of hemicellulose and AIL, this study indicated that the presence of PEG 6000 had little influence on compositional changes of SCB (Nasirpour et al., 2014; Sindhu et al., 2012; Wang et al., 2020a). This may be attributed to differences among intrinsic properties in materials and pretreatment technologies and conditions.

Compositions of sugars and inhibitors in liquors undergoing PEG 6000-assisted sulfuric acid pretreatment are summarized in Table 2. As shown, addition of PEG 6000 altered the compositions of prehydrolysates slightly. The main changes observed in the prehydrolysate composition involved an increase in total xylose proportion from 18.57 g to 20.87 g per 100 g raw biomass when H₂SO₄ concentrations ranged from 1% to 3%. The total glucose yields increased gradually from 1.92 g to 2.53 g/100 g raw material, which was much lower than the level seen for xylose, indicating that hemicellulose degradation was easier than cellulose degradation, and this proposed acid condition was favored for the retention of cellulose. In addition, other degradation products, such as furfural formed by dehydration of xylose, HMF formed from glucose, and acetic acid released from acetyl groups in the hemicelluloses, were also identified and are described in Table 2. The yields of inhibitors were all increased with harsher acid conditions. Only small amounts of HMF (0.02–0.04 g/L) were detected in the prehydrolysate, and this was ascribed to limited decomposition of cellulose and a faster rate for degradation of cellulose to glucose than for degradation of glucose to HMF (Rajan and Carrier, 2014). The increasing proportions of furfural (0.11–0.63 g/L) and acetic acid (3.57–4.10 g/L) further

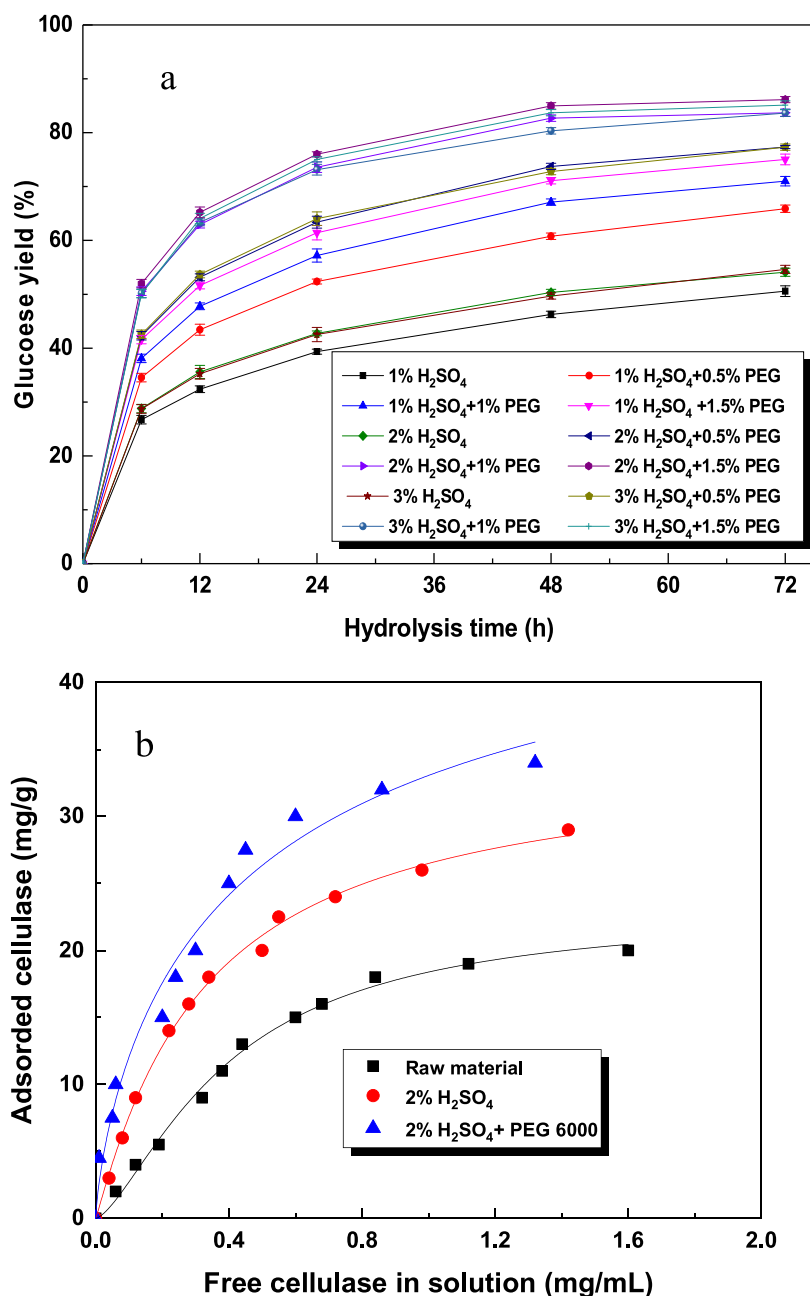


Fig. 2. Glucose yield after PEG 6000 assisted H₂SO₄ pretreatment with various concentration (a) and cellulase adsorption isotherm on different samples (b).

implied that the dissolved hemicellulose was more easily dehydrated to xylose and more readily lost acetyl groups under increasingly acidic conditions. The addition to PEG 6000 exhibited a slight influence on structural and prehydrolysate compositions, which was in accordance with previous literature describing the use of Tween 20-assisted dilute acid pretreatment of wheat straw (Qi et al., 2010b). In summary, the material balance for 1.5 % PEG 6000-assisted 2% H₂SO₄ pretreatment and subsequent enzymatic hydrolysis was calculated to yield approximately 69.46 g of dry pretreated solids, including 41.78 g of glucan, 4.70 g of xylan and 20.53 g of AIL, collected from 100 g raw SCB. After pretreatment, approximately 20.88 g of xylose, 2.20 g of glucose, 0.71 g of lignin, 3.80 g of acetic acid, 0.03 g of HMF and 0.33 g of furfural were obtained in the prehydrolysate. After enzymatic saccharification, approximately 42.31 g of glucose together with 5.01 g of xylose were produced in the enzymatic hydrolysate, which accounted for 91.15 % of the glucan and 93.82 % of the xylan in the pretreated SCB.

3.3. Effect of PEG 6000 assisted H₂SO₄ pretreatment on enzymatic hydrolysis

The impact of PEG 6000 and H₂SO₄ on the glucose yield from enzymatic digestion is depicted in Fig. 2a. As shown, without the addition of PEG 6000 during H₂SO₄ pretreatment, when the acid concentration was enhanced from 1% to 3%, the glucose yields increased from 50.58 % to 54.62 %, an increase of 7.98 % in glucose yield; this suggested that it is impracticable to substantially promote enzymatic hydrolysis by increasing the H₂SO₄ dosage alone. Hence, PEG 6000 was added to the H₂SO₄ pretreatment to enhance the glucose yield. For substrates treated with 0.5 %, 1.0 % and 1.5 % PEG 6000-assisted 1% H₂SO₄, the glucose yields after hydrolysis for 72 h were 65.88 %, 71.01 % and 75.04 % (samples 2–4), corresponding to increases of 30.24 %, 40.36 % and 48.34 %, respectively, in glucose yields over those seen for pretreatment with 1% H₂SO₄ alone (50.58 %). As the H₂SO₄

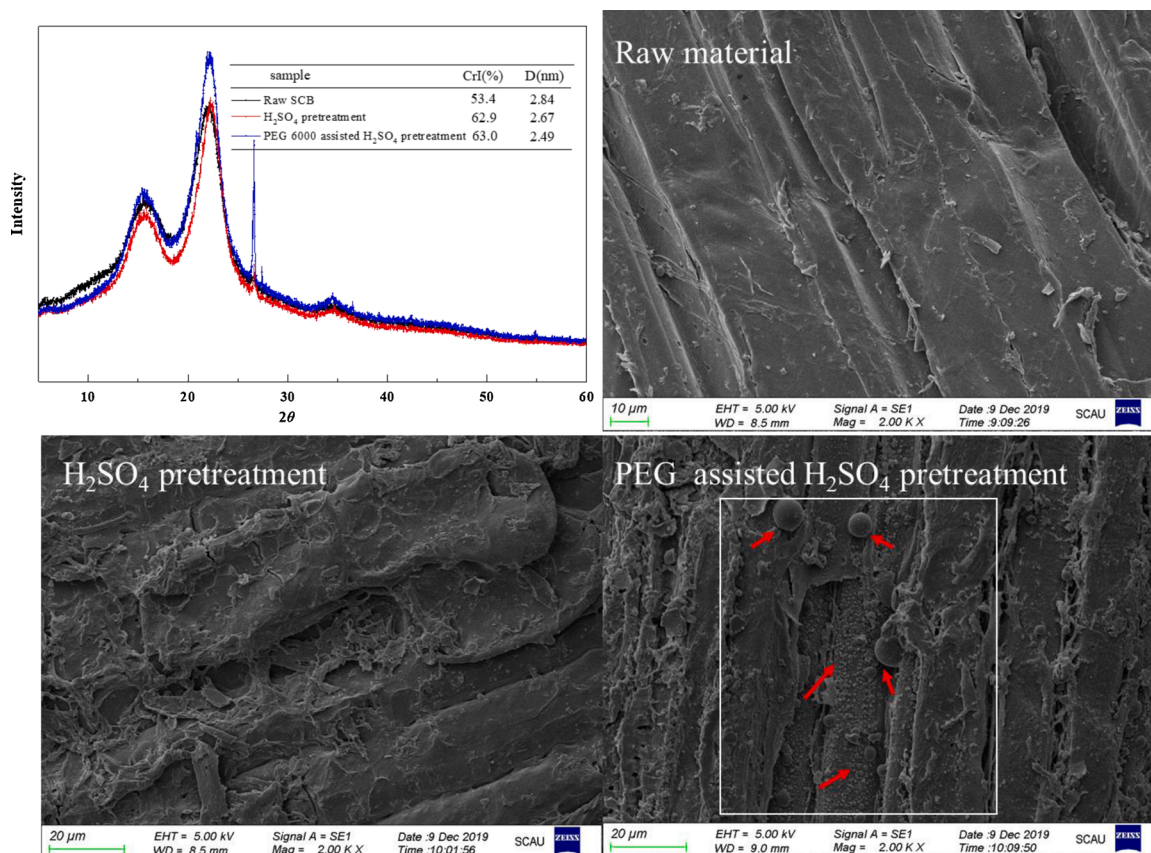


Fig. 3. Characterization of native and pretreated SCB of XRD and SEM.

concentration was increased to 2%, various concentrations of PEG 6000 resulted in 72 h glucose yields of 54.09 %, 77.34 %, 83.71 %, 86.14 % (samples 6–8), respectively, with increases of 42.96 %, 54.74 %, 59.23 % compared with 2% H₂SO₄ pretreatment alone (54.09 %). The combination of PEG 6000 and lignin allowed higher cellulase accessibility to cellulose and hemicellulose (Kim et al., 2007).

The same trend was also reflected in different concentrations of PEG 6000-assisted 3% acid pretreatment (samples 9–12), where significant increases of approximately 41.51 %, 53.13 % and 55.82 % in glucose yield were found compared with that for 3% acid pretreatment alone. However, as the H₂SO₄ concentration was raised to 3%, the glucose yields obtained from enzymatic hydrolysis declined slightly compared with those obtained with 2% H₂SO₄ pretreatment at the same PEG 6000 concentrations. This could be ascribed to the lower retention of glucan under severe pretreatment conditions. Therefore, the highest glucose yield of 86.14 % was achieved after 72 h hydrolysis of SCB with 1.5 % PEG 6000 combined with 2% H₂SO₄, which resulted in a 91.15 % yield for glucose from pretreated SCB. Although addition of surfactant had a slight influence on structural composition, there were significant increases in glucose yield, which could be attributed to improved enzymatic activity, decreased lignin adsorption of cellulase enzymes by penetration of plant cell walls and surface modifications of lignin, which was in accordance with previous studies (Qi et al., 2010a; Sun et al., 2020). A higher glucose concentration of 8.2 g/L with Tween-assisted acid pretreated wheat straw was achieved by (Qi et al., 2010a). Chen et al. found that Tween-assisted FeCl₃ pretreatment enhanced cellulase hydrolysis of Eucalyptus waste fibers to 91.3 % by eluting enzymatic inhibitors with Tween (Chen and Fu, 2013).

It is necessary to evaluate the overall glucose (a) and xylose (b) recovery yield resulting during the process of pretreatment and enzymatic hydrolysis to determine further biofuel production. As shown in Fig. S1, more glucose and xylose were recovered by adding more PEG 6000 into

pretreatments with the same H₂SO₄ concentration. Specifically, since most glucose was recovered in the enzymatic digestibility stage, the highest overall glucose yield of 44.51 g/100 g raw material (including pretreatment and enzymatic saccharification) was acquired from SCB pretreated with 1.5 % PEG 6000-assisted 2% H₂SO₄, reaching 90.62 % of the theoretic maximum. A large number of xyloses was achieved from prehydrolysates in the acidic environment, and the highest total xylose yield of 25.89 g/100 g raw SCB was obtained under the same pretreatment conditions (consist with glucose recovery), representing 95.41 % of the total xylose in the untreated SCB.

Moreover, cellulase adsorption isotherms for native, 2% H₂SO₄ and PEG 6000-assisted 2% H₂SO₄ pretreated SCB were determined to address the effect of PEG 6000 assisted pretreatment on cellulase accessibility. After fitting the data to the Langmuir isotherm, the corresponding parameters were obtained for the different samples. As shown in Fig. 2b and Table S1, PEG 6000 plus 2% H₂SO₄ exhibited higher adsorption capacity ($E_{\max} = 52.41$ mg/g) than the untreated substrate ($E_{\max} = 23.06$ mg/g) and 2% H₂SO₄ pretreated SCB ($E_{\max} = 34.49$ mg/g). The measurement of binding strength ($K_R = K_d \cdot E_{\max}$) helps in analyzing the affinities of enzymes for substrates. A significant increase in the value of K_R after combined PEG 6000 + 2% H₂SO₄ pretreatment (142.03 L/g cellulase) and standalone dilute acid pretreatment (114.17 L/g cellulase) of SCB showed that cellulase had a greater affinity for pretreated biomass than it did for native biomass (K_d 90.63 L/g cellulase). Increased biomass accessibility for cellulase with SCB pretreated with PEG 6000 and 2% H₂SO₄ helped increase the efficiency of enzymatic hydrolysis.

3.4. Characterization of native and pretreated SCB by XRD, FTIR and SEM

To determine why PEG 6000 presented a considerable effect on

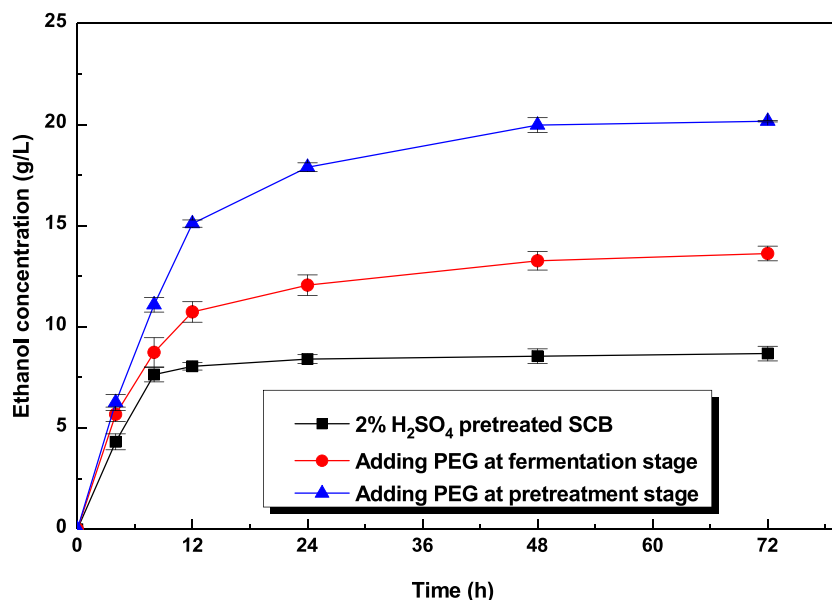


Fig. 4. The ethanol concentration of dilute acid pretreated SCB (10 % solids loading) with addition of 1.5 % PEG 6000 at different stage.

enzymatic hydrolysis during H₂SO₄ pretreatment SCB, structural features of native and pretreated SCB were examined by using XRD, FTIR and SEM. After pretreatment, cellulosic crystallinity was changed by reducing the crystal hydrogen bonding and removing of amorphous components (Peculyte et al., 2015). XRD data and CrI values for raw and H₂SO₄ pretreated SCB (without PEG and with 1.5 % PEG) are described in Fig. 3. Compared with an initial CrI value of 53.4 % in raw SCB, a higher CrI of 62.9 % was obtained after 2% H₂SO₄ pretreatment, and this was ascribed to degradation of amorphous hemicellulose under dilute acid conditions, which is verified in Table 1. However, the crystallinity showed little change between substrates pretreated with and without PEG 6000 (~63.0 %), which was in accord with the composition analysis contained in Table 1. This result was in consistent with the FeCl₃ pretreated substrates with and without Tween 80 proposed by (Chen and Fu, 2013). Furthermore, cellulose crystallite sizes of different samples were also calculated and are shown in Fig. 3. The average size of crystallites in untreated SCB was 2.84 nm (002), and the crystallite size decreased to 2.67 nm after H₂SO₄ pretreatment. A smaller size of 2.49 nm was observed after adding PEG 6000 in H₂SO₄ pretreatment, suggesting that pretreatments with either H₂SO₄ or PEG 6000-assisted H₂SO₄ decreased crystallite sizes, which facilitated enzymatic hydrolysis and increased the accessible surface area (Sun et al., 2015).

The chemical structures of raw and pretreated SCB (without PEG and with 1.5 % PEG) were analyzed using FTIR. The FTIR spectra of SCB were mainly comprised of representative spectra for cellulose, hemicellulose and lignin. As shown in Fig. S2, the band intensities at 1734 cm⁻¹, which are assigned to carbonyls or acetyl groups in hemicellulose, decreased dramatically after pretreatment of raw SCB, which indicated that most of the hemicellulose was removed. However, lignin (1513 cm⁻¹ and 1604 cm⁻¹) and cellulose (897 cm⁻¹) peaks showed no significant changes, suggesting that dilute acid under mild conditions did not effectively remove lignin but better retained cellulose components. As expected, there were no significant differences in the functional groups of cellulose, hemicellulose or lignin resulting from acid and PEG 6000-assisted acid pretreatments. This resulted because the addition of PEG 6000 caused no changes in the bonding of lignocellulosic substrates (Li et al., 2019).

The ultrastructure and surface features of raw SCB samples and those pretreated with sulfuric acid and PEG 6000-assisted sulfuric acid were observed by SEM. As shown in Fig. 3, the raw substrate exhibited an intact and compact morphology due to orderly arrangements of

cellulose, hemicellulose and lignin. After H₂SO₄ pretreatment, numerous cracks, fragments and gullies were observed in the originally smooth surface. The formation of a rough and loose structure resulted from hemicellulose removal under acidic conditions. However, with PEG 6000-assisted H₂SO₄ pretreatment, some smaller fragments and looser structures were produced, which was consistent with the decreased crystallite sizes seen after PEG 6000-assisted H₂SO₄. PEG 6000 was found to attach to surfaces or pores of pretreated SCB and form spherical structures with different sizes due to H-bonds and hydrophobic interactions with lignin components (Börjesson et al., 2007). Attached PEG 6000 lubricated the combination of cellulose with cellulase by binding with lignin (Qing et al., 2010). Therefore, the XRD, FTIR and SEM results implied that the promotion of enzymatic digestibility caused by adding PEG 6000 could be attributed to increased access for cellulase resulting from smaller crystalline sizes and reductions in nonproductive adsorption to lignin.

3.5. SSF of the pretreated SCB to produce ethanol

SSF experiments were conducted by using SCB pretreated with 1.5 % PEG 6000-assisted 2% H₂SO₄ to explore effects on ethanol production. Additionally, pure 2% H₂SO₄ pretreated substrate was also used as a control, and the results are depicted in Fig. 4. During all SSF processes, there were similar trends for change in ethanol concentration, which increased rapidly in the initial 12 h and then rose slowly to achieve a steady state at 72 h. In addition, the fermentation rate for the combined pretreatment was 1.26 g/L/h for 12 h, which was greater than that seen for the control, 0.67 g/L/h. After fermentation for 72 h, final ethanol concentrations of 20.17 g/L (58.19 %) and 8.68 g/L (25.06 %) were produced. The low ethanol concentration for the control sample might be attributable to nonproductive adsorption of cellulase on lignin as described in Section 3.4. In addition, our previous research shown that adding surfactants in the fermentation stage instead of in the pretreatment stage also enhanced ethanol production (Fan et al., 2019). Hence, the addition of adding PEG 6000 into both the pretreatment and fermentation stages was also studied. Compared to the control, addition of PEG 6000 to the fermentation stage also accelerated the fermentation rate, especially in the first 12 h, and a fermentation rate of 0.89 g/L/h was observed. After 72 h, approximately 13.63 g/L (39.35 %) of ethanol was achieved, which was higher than the yield seen for the control but lower than that seen by adding PEG 6000 during the pretreatment stage.

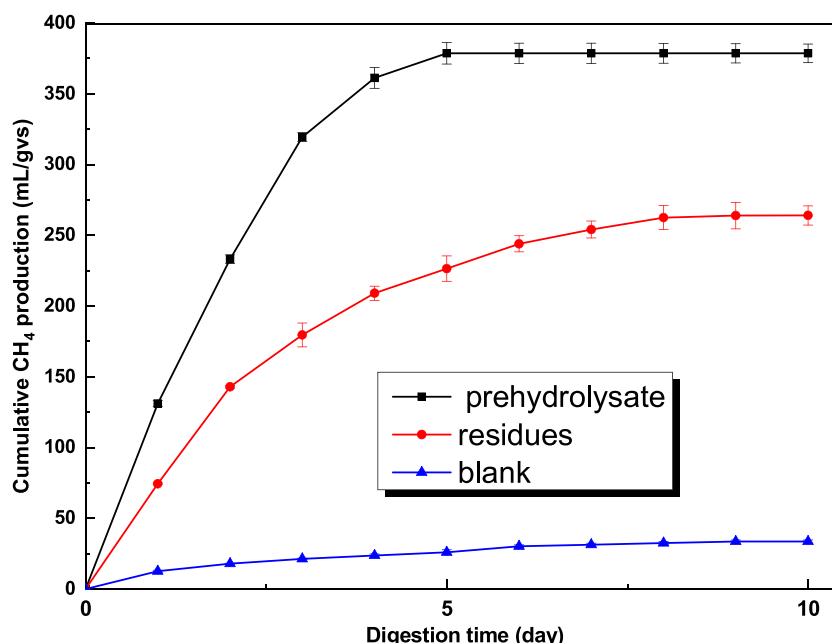


Fig. 5. Cumulative methane production from prehydrolysate and stillage residue.

Note: The experiment of biomethane production was continuously conducted until day 30, the cumulative CH₄ production remained constant from day 9.

This indicated that addition of PEG 6000 during the pretreatment stage yielded superior performance in boosting ethanol production (Qi et al., 2010a). The positive effect of PEG 6000 addition in the pretreatment stage on ethanol production resulted from degradation of the lignocellulose structure to allow enzyme accessibility at a higher pretreatment temperature.

3.6. Biomethane production from pretreatment liquor and fermentation residues

After ethanol production, high xylose content in the pretreatment liquor and residual sugars in the stillage are easily fermented and utilized as the inoculum in anaerobic digestion (Narra et al., 2018). As shown in Table 3, the prehydrolysate had 3.61 % TS, 3.73 % VS, 5600 mg/L COD and a pH of 0.71. To avoid acidification of the reaction system, the hydrolysate was neutralized with 5 M Na₂CO₃ to establish a pH in the range of 6.50–7.00. After ethanol recovery, the residues had 19.11 % TS, 18.38 % VS and a pH of 4.50. Results for biomethane accumulation over 10 days are described in Fig. 5. The maximum biomethane yield of 378.65 mL/g VS was achieved from prehydrolysate, which was higher than that obtained from SCB stillage (264.05 mL/g VS). A higher production rate and shorter fermentation time were observed for the prehydrolysate, because xylose was directly fermented by the inoculum.

In addition, during the anaerobic digestion process, changes in some parameters, such as pH, NH₃-N, COD, and VS, influenced the reaction. Specifically, the values of pH (6.80–7.50) and NH₃-N (500–900 mg/L) fluctuated within certain ranges, revealing that there was no acidification or suppression of ammonia nitrogen formation (Wang et al., 2018b). Final COD removal efficiencies of 87.62 % and 83.59 % and VS degradation yield of 91.22 % and 88.72 % were determined for prehydrolysate and residues, respectively, indicating that most organic substrates were converted to biomethane (Suanon et al., 2017). Thus, anaerobic digestion of the stillage and prehydrolysate for biogas production was feasible, and the entire experimental process investigated provided a reference for large-scale production.

4. Conclusion

LHW, ethanol, dilute acid and dilute alkali pretreatments of SCB were performed in the presence of three different surfactants (PEG 6000, Triton X-100 and Tween 80). It was established that PEG 6000-assisted H₂SO₄ was favorable for production of fermentable sugars during the enzymatic and fermentative reactions. Maximum enzymatic hydrolysis of 86.14 % and bioethanol concentration of 20.16 g/L were obtained for PEG 6000-assisted 2 % H₂SO₄ pretreatment, which were increased by 59.24 % and 132 % compared to those seen for pretreatment with 2 % H₂SO₄ alone. Effects of PEG 6000 on cellulose conversion during acid pretreatment were due to formation of smaller crystalline fragments and PEG 6000 adherence to the SCB surface after combined pretreatment of SCB. Furthermore, anaerobic digestion of prehydrolysate and stillage residue for biomethane production provided an environmentally friendly technology for achieving higher energy efficiency. Future experiments will be necessary to analyze the mechanism of the surfactant effect on enzymatic hydrolysis and fermentation and provide an economic analysis.

CRediT authorship contribution statement

Meishan Fan: Writing - original draft, Methodology, Data curation. **Chenbiao Zhao:** Formal analysis, Methodology. **Xinyu Huang:** Formal analysis, Methodology. **Hongdan Zhang:** Supervision, Validation, Data curation. **Jun Xie:** Supervision, Project administration, Writing - review & editing.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary material related to this article can be found, in the online version, at doi:<https://doi.org/10.1016/j.indcrop.2021.114006>.

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